Learning Spectral Geometry for Kernel Regression

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Abstract

Gradient-boosted trees provide a strong benchmark for supervised learning on tabular data. We investigate kernel regression as an alternative in which the representation of similarity is learned separately from the final prediction coefficients. The proposed model combines learned Fourier coordinates, a positive-definite metric and a Laplace kernel. A canonical parameterization removes compensating amplitude, metric and bandwidth scales: the metric has unit trace, amplitudes are derived from its blocks and a symmetric trace-zero matrix parameterizes its shape through a normalized matrix exponential. This identification is algebraic and does not extend to identification of the whole model from finite observations. Geometry is learned from supervised prediction loss and followed by kernel ridge regression. Categorical contrasts extend the representation to mixed inputs, and binary-label regression with separate calibration supports classification. Each dataset uses a single spectral bank and a single ridge model. Against tuned gradient-boosted trees, the spectral model has lower test error on California Housing and on a chronological holdout of hourly Bike Sharing. On Taiwan credit default it has slightly higher ranking accuracy and nearly equal calibrated log loss, while on Adult income the boosted trees have higher ranking accuracy and lower log loss. Comparisons of approximate and exact ridge fits show that decoder accuracy and geometry refinement both affect performance. Tuning budgets differ between model families, the California holdouts were reused during development, and dense matrix exponentiation and exact ridge solves carry cubic costs. Transfer across tables is not evaluated.

Keywords: Kernel ridge regression; spectral representation learning; metric learning; Laplace kernels; tabular learning; identifiable parameterization.

1 Introduction

Gradient-boosted decision trees (GBDTs) have long been a leading practical choice for supervised learning on tabular data. Systems such as XGBoost [3], LightGBM [7] and CatBoost [10] combine strong predictive performance with efficient implementations. Their practical appeal extends to large datasets: scalable tree-boosting systems are designed to handle substantial training volumes efficiently [3, 7]. Heterogeneous features, irrelevant inputs and irregular response functions make tabular prediction challenging across data scales. A benchmark of medium-sized tables, with about 10,000 training samples, found tree-based models ahead of the deep learning models it evaluated [5].

Tabular foundation models reuse a pretrained neural model across prediction tasks, taking a new table's labeled examples as context. TabPFN demonstrates this approach through pretraining on synthetic datasets and prediction on new datasets using in-context learning [6]. This approach makes transfer across tables an explicit training objective.

Conventional boosted trees are fitted to individual prediction tasks. Their split thresholds and leaf values are fitted to a particular table, its columns and its labels. Each tree partitions that

feature space into regions with constant predictions, and the ensemble combines those partitions into a task-specific geometry. A model trained on one schema does not by itself supply a rule for comparing observations in a new table with different columns or meanings. Tree-based transfer requires additional structure beyond that conventional training setup. This motivates studying representations in which the notion of similarity can be expressed and learned separately from the final regression fit.

We explore kernel regression as such an alternative. A kernel assigns a similarity to two observations, and the regression model uses those similarities to predict from labeled examples. With a fixed kernel, ridge regression reduces to a regularized linear solve. Learning can therefore be organized into two parts: learning how observations should be compared, and fitting the coefficients that turn those comparisons into predictions. The present paper studies this separation within individual datasets. Reusing the learned representation across tables would require training across tasks and a way to handle different feature schemas.

Learning similarity requires deciding which input differences matter for prediction. Euclidean distance on standardized columns assigns the same distance to a one-unit change wherever it occurs. In housing data, however, the relevance of a change in income, occupancy or location can depend on both its scale and the other attributes of the observation. We build a similarity in three stages: represent variation at several scales, learn which directions matter, and determine how distance controls the influence of training examples.

Variation at several scales is represented through Fourier coordinates. Each numerical input is mapped to sine–cosine pairs with learned frequencies. Low frequencies describe gradual variation; higher frequencies distinguish changes over shorter input intervals. Using several frequencies lets a feature participate at several scales without assigning a separate region to every observed threshold. Fourier coordinates also connect this construction to the spectral representation of stationary kernels. Random Fourier features use that connection to approximate a specified kernel [11], while spectral mixture kernels learn a parametric spectrum [19]. Here the frequencies are optimized for the regression task. Their use as basis functions does not assume a periodic target, although a finite Fourier map can alias distant inputs, which matters when extrapolating.

A learned metric then determines distances in Fourier space. Frequencies alone do not say which encoded differences are relevant to the target. A metric stretches some directions and contracts others. Its cross terms allow combinations of features and frequencies to affect distance together. For example, it can distinguish a direction combining changes in location and income from either change considered separately. We also recover feature masses and relative frequency amplitudes from this metric, so that the allocation of weight has a defined mathematical meaning. A full metric can make the resulting kernel nonstationary in the original inputs; the spectral representation remains useful without requiring the final similarity to depend only on input differences.

A Laplace kernel converts the resulting distance into similarity. Similarity decays exponentially with distance, and a learned bandwidth determines how local the comparison is. This gives a positive-semidefinite kernel after the learned transformation [13, Theorem 1 and Corollary 1]. Its linear dependence on distance in the exponent permits a sharper change near coincident observations than the squared distance used in a Gaussian kernel. This makes it a reasonable candidate for tabular responses with rapid local variation. Applying it to a joint distance yields a nonadditive similarity, even when the squared distance is a sum of featurewise contributions.

Together, the components give

$$k_\theta(x, x') = \exp\left[-\frac{\sqrt{(u_\omega(x) - u_\omega(x'))^\top B(u_\omega(x) - u_\omega(x'))}}{T}\right]. \tag{1}$$

Here u_ω is the spectral representation, B is the learned metric and T is the bandwidth. Kernel ridge regression then uses this similarity in an expansion over training observations, with coefficients fitted simultaneously by a regularized solve. This construction belongs to the broader family of learned representations composed with kernels [20].

There is a direct parallel with cross-attention: a new observation supplies a query, training observations supply keys and label information supplies values. Both constructions compare the query with the available examples and aggregate value information. Normalizing our Laplace similarities across training examples produces a kernel smoother in softmax cross-attention form. Ridge regression instead adjusts the labels through a solve involving similarities among the training examples. Section 5.3 makes the two readouts precise; the experiments use the ridge readout.

The representation and distance parameters contain redundant scales. Increasing an amplitude and reducing the corresponding metric entries can leave the kernel unchanged. Multiplying the metric by a positive constant can likewise be offset by multiplying the bandwidth by its square root. We remove these redundancies by assigning the metric unit trace, deriving amplitudes and feature masses from its blocks and parameterizing its positive-definite interior through a symmetric trace-zero matrix exponential. The resulting coordinates distinguish representation scale, metric shape and bandwidth without adding compensating parameters.

A single bank here means one Fourier map, one metric and one bandwidth, followed by one ridge fit. We evaluate this construction with four frequencies per feature on California Housing. Across three matched random splits, the refined model with exact ridge decoding achieves mean test root mean squared error (RMSE) of \$42,366, compared with \$43,186 for tuned CatBoost and \$43,568 for tuned LightGBM. Separate comparisons measure the changes from replacing the approximate decoder and from further geometry training. The holdouts were reused during development and the tuning budgets differ between families. The additional experiments retain one kernel per dataset and extend the representation to categorical predictors. They compare hourly Bike Sharing counts, Taiwan default labels and Adult income labels with tuned CatBoost under fixed selection protocols. No cross-task pretraining is evaluated.

2 From frequencies to a distance

2.1 The spectral starting point

For a continuous real stationary positive-semidefinite kernel with $k(0) = 1$, Bochner's theorem gives a symmetric probability measure μ such that

$$k(x - x') = \int_{\mathbb{R}^d} \cos\big(2\pi\xi^\top (x - x')\big)\, d\mu(\xi). \tag{2}$$

We use frequency in cycles per input unit, hence the factor 2π. A standard account is given by Rasmussen and Williams [12, Theorem 4.1], who state the theorem in complex-exponential form; for a real symmetric kernel it reduces to (2). Equation (2) connects a kernel's notion of similarity to the frequencies it contains.

Let $x \in \mathbb{R}^d$ denote an input after a fixed, training-fitted transformation. For feature j, choose K positive ordered frequencies and form

$$u_{jk}(x) = \begin{bmatrix} \cos(2\pi\omega_{jk} x_j) \\ \sin(2\pi\omega_{jk} x_j) \end{bmatrix}, \qquad u_\omega(x) = \operatorname{col}_{j,k} u_{jk}(x) \in \mathbb{R}^p, \qquad p = 2dK. \tag{3}$$

Each pair lies on the unit circle, and its angle changes by $2\pi\omega_{jk}$ per unit change in x_j. For pair

weights $q_{jk} \geq 0$, the elementary identity

$$\sum_{j,k} q_{jk} \|u_{jk}(x) - u_{jk}(x')\|^2 = 2 \sum_{j,k} q_{jk} \{1 - \cos[2\pi\omega_{jk}(x_j - x'_j)]\} \tag{4}$$

makes the induced stationary distance explicit. With weights summing to one, the corresponding weighted inner product is the atomic version of (2).

We learn the frequencies from labels. To remove sign and permutation conventions within each feature, set

$$\omega_{jk} = \sum_{\ell=1}^{k} \exp(\nu_{j\ell}), \qquad k = 1, \ldots, K. \tag{5}$$

There is no additional learnable input-frequency multiplier. The coordinate origin is fixed by preprocessing, and no phase parameters are introduced. These choices remove redundant descriptions; they do not prevent different frequencies from agreeing on a finite sample.

2.2 Metric shape and bandwidth

Write $\Delta u = u_\omega(x) - u_\omega(x')$. The model is

$$d_B(x, x')^2 = \Delta u^\top B \Delta u, \qquad k_\theta(x, x') = \exp\{-d_B(x, x')/T\}, \quad B \succeq 0, \quad \operatorname{tr} B = 1, \quad T > 0. \tag{6}$$

The eigenvectors of B select directions in Fourier space; its eigenvalues determine their relative stretching. The bandwidth T controls the overall distance scale.

The Euclidean Laplace kernel is positive semidefinite in any dimension [13, Theorem 1 and Corollary 1], and composing it with a feature map preserves that property. Specifically, choose R with $RR^\top = B$ and let $z(x) = R^\top u_\omega(x)$. Equation (6) is $\exp(-\|z(x) - z(x')\|/T)$, so every Gram matrix (the matrix of pairwise kernel evaluations) is positive semidefinite and $k_\theta(x, x) = 1$.

The Fourier representation does not make the final model stationary automatically. A block-diagonal metric with an isotropic block for each sine–cosine pair gives the difference-only expression in (4). A general full metric can also respond to the individual input locations, through unequal within-pair stretching and cross-pair terms. Thus the frequencies in (3) describe the representation; they are not, in general, the spectral measure of the final kernel.

3 Removing redundant scales

Suppose a preliminary description has an amplitude matrix D_0, a metric M and bandwidth T_0. Its squared scaled distance is

$$\frac{\Delta u^\top D_0 M D_0 \Delta u}{T_0^2} = \Delta u^\top H \Delta u, \qquad H = \frac{D_0 M D_0}{T_0^2}. \tag{7}$$

Only H enters the kernel, so independent amplitude, metric and bandwidth parameters admit equivalent descriptions of the same distance.

Proposition 1. *For any nonzero positive-semidefinite H, the unique decomposition $H = B/T^2$ with $\operatorname{tr} B = 1$ and $T > 0$ is*

$$B = H/\operatorname{tr} H, \qquad T = (\operatorname{tr} H)^{-1/2}. \tag{8}$$

Proof. Taking traces gives $\operatorname{tr} H = 1/T^2$, which determines T and then B. □

3.1 Recovering amplitude and importance

Let $B_{jk,jk}$ be the 2×2 principal block for pair (j, k). Define

$$w_{jk} = \operatorname{tr} B_{jk,jk}, \qquad \rho_j = \sum_k w_{jk}, \qquad a_{jk}^2 = \frac{w_{jk}}{\rho_j}. \tag{9}$$

Then $\sum_j \rho_j = 1$ and $\sum_k a_{jk}^2 = 1$ whenever $\rho_j > 0$. The feature mass ρ_j records how much of the trace is assigned to feature j; a_{jk}^2 divides that mass among its frequencies. These are learned through B, rather than fitted as extra scales.

For positive pair masses, let D have blocks $\sqrt{w_{jk}/2}\, I_2$. The remaining shape is

$$\Gamma = D^{-1} B D^{-1}, \qquad B = D \Gamma D, \qquad \operatorname{tr} \Gamma_{jk,jk} = 2. \tag{10}$$

This is a unique decomposition under the stated block normalization. It separates allocation of mass from within-pair anisotropy and cross-pair coupling. If a pair mass is zero, this inverse is undefined; the zero-mass coordinates must be removed or treated as a boundary case. The positive-definite implementation below avoids that boundary.

Feature mass alone is not predictive importance. A small mass at a high frequency can still produce substantial local sensitivity, and correlated inputs complicate attribution.

3.2 Unique coordinates for the positive-definite interior

Use

$$B(A) = \frac{\exp(A)}{\operatorname{tr} \exp(A)}, \qquad A = A^\top, \quad \operatorname{tr} A = 0, \qquad T = \exp(\tau). \tag{11}$$

Matrix exponential methods for positive-definite trace-one matrices are established [15]. Here the trace-zero restriction matters: without it, A and $A + cI$ would give the same B.

Proposition 2. *Equation (11) is a bijection between symmetric trace-zero matrices and positive-definite matrices of trace one. Its inverse is*

$$A = \log B - \frac{\operatorname{tr}(\log B)}{p} I. \tag{12}$$

Proof. The exponential of a real symmetric matrix is positive definite. Normalizing its trace gives an admissible B. Conversely, the symmetric logarithm of a positive-definite B exists uniquely. Subtracting its mean eigenvalue gives trace zero, and exponentiating recovers a positive scalar multiple of B. Trace normalization removes that scalar. Applying the same logarithm to any preimage gives (12). □

We optimize $p(p+1)/2 - 1$ coefficients in an orthonormal basis of symmetric trace-zero matrices. This avoids retaining an unused trace direction inside the optimizer. Frequency signs, within-feature ordering, amplitude scales, metric scale and matrix-generator shifts have now been fixed. Global identifiability of (ω, B, T) from observations is a separate question: finite-sample aliasing, constant inputs and observationally equivalent representations can remain.

4 What the geometry can express

For a small displacement h,

$$d_B(x, x+h)^2 = h^\top J_u(x)^\top B J_u(x) h + o(\|h\|^2), \tag{13}$$

where J_u is the Fourier-map Jacobian. Thus $J_u(x)^\top B J_u(x)$ is the local metric in input coordinates. In the simple case $B_{jk,jk} = (w_{jk}/2)I_2$ with zero cross blocks, its jth diagonal entry is

$$I_j = \frac{1}{2}\sum_k w_{jk}(2\pi\omega_{jk})^2. \tag{14}$$

Frequency and mass jointly determine sensitivity. A full B adds cross-feature terms and allows sensitivity to vary with location.

The Laplace readout also makes the kernel nonadditive. Even if d_B^2 is a sum of featurewise distances, applying $\exp(-\sqrt{\cdot}/T)$ to that sum generally cannot be written as a sum of separate feature kernels. Consequently, a ridge expansion can express interactions. The bandwidth controls how that joint distance becomes similarity; it is not a separate interaction-order parameter.

The Euclidean Laplace kernel has a cusp at zero distance, whereas the squared-exponential kernel is smooth there. Their stationary spectra have different tails: under angular-frequency convention, the Laplace spectral density in p dimensions is proportional to $(T^{-2} + \|\xi\|^2)^{-(p+1)/2}$, while the Gaussian spectrum decays exponentially in squared frequency [12, Section 4.2.1, eq. (4.15) with $\nu = 1/2$]. The experiments do not include a comparison with a Gaussian readout.

5 Learning and regression

5.1 Learning a geometry with support and query sets

Geometry parameters are trained using labeled support and query episodes. With disjoint sets S, Q, normalized targets $\widetilde{y}$, and support mean $\overline{y}_S$, predict

$$\widehat{\widetilde{y}}_q^{\,\mathrm{ep}} = \overline{y}_S + k_{qS}(G_{SS} + \eta I)^{-1}(\widetilde{y}_S - \overline{y}_S \mathbf{1}), \tag{15}$$

$$\mathcal{L} = \frac{1}{|Q|}\sum_{q\in Q}(\widehat{\widetilde{y}}_q^{\,\mathrm{ep}} - \widetilde{y}_q)^2 + \frac{10^{-4}}{p}\|A\|_F^2. \tag{16}$$

Here k_{qS} is a row of kernel evaluations and $\eta > 0$ is learned. Gradients pass through the support solve, Fourier map and matrix exponential. Query error rewards a geometry that supports prediction beyond the rows used in that episode's solve. All labels used for this stage belong to the designated fitting pool.

5.2 Exact regression after geometry selection

For fixed geometry and $\lambda > 0$, fit an unpenalized intercept and a regularized function in the reproducing kernel Hilbert space (RKHS) of k_θ:

$$\min_{b\in\mathbb{R},\, f\in\mathcal{H}_{k_\theta}} \frac{1}{n}\sum_i (\widetilde{y}_i - b - f(x_i))^2 + \lambda\|f\|_{\mathcal{H}_{k_\theta}}^2. \tag{17}$$

The semiparametric representer theorem [14, Theorem 2], with the constant function as the unpenalized component, gives a finite kernel expansion plus the intercept. For $G_{ij} = k_\theta(x_i, x_j)$ and

$C_\lambda = G + n\lambda I$, its solution is

$$b = \frac{\mathbf{1}^\top C_\lambda^{-1} \widetilde{y}}{\mathbf{1}^\top C_\lambda^{-1} \mathbf{1}}, \qquad \alpha = C_\lambda^{-1}(\widetilde{y} - b\mathbf{1}), \qquad \mathbf{1}^\top \alpha = 0. \tag{18}$$

Two right-hand sides share one Cholesky factorization; we never form the inverse explicitly. In original target units,

$$\widehat{y}(x) = \mu_y + \sigma_y \left[b + \sum_i \alpha_i k_\theta(x, x_i) \right]. \tag{19}$$

The final free intercept differs from the support mean in episodic training. Likewise, η is the episodic diagonal shift, whereas $n\lambda$ is the shift associated with the final mean-loss objective. They are not interchangeable tuning parameters.

Nyström approximation reduces the cost by representing the kernel through a subset of training observations, called landmarks [18]. We use it during checkpoint selection and compare it with exact decoding. Fourier dimension p is not a rank bound for the Laplace Gram matrix: the nonlinear readout can produce a full-rank Gram matrix even from a compact embedding. A small Fourier representation therefore does not guarantee that a small landmark set preserves its predictive accuracy.

Table 1: Fitting one spectral kernel. The final ridge coefficients are fitted after geometry selection.

Stage	Operation
Preprocessing	Fit input and target transformations on the inner-training rows.
Geometry training	Draw disjoint support and query rows; compute Fourier coordinates and the normalized matrix exponential; differentiate query error through the support solve.
Validation	Select a geometry checkpoint, the final ridge penalty and optional output clipping using validation predictions.
Final fit	Freeze these choices, refit geometry for the selected number of steps on the outer-training pool, then fit the exact ridge head.
Prediction	Cache the selected metric transform and training embeddings; evaluate query similarities and apply the fitted coefficients.

5.3 The connection to cross-attention

Scaled dot-product attention compares query representations with key representations from a context set and uses the resulting weights to combine the associated values [16, Section 3.2.1]:

$$\mathrm{Att}(\mathsf{Q}, \mathsf{K}, \mathsf{V}) = \mathrm{softmax}_{\mathrm{row}}\left(\frac{\mathsf{Q}\mathsf{K}^\top}{\sqrt{d_k}} \right) \mathsf{V}, \tag{20}$$

where d_k is the key dimension. In cross-attention the queries and the context come from different sets; here the queries are test observations X_* and the context is the labeled training set X. The sans-serif matrices are representations, distinct from the episode sets S, Q and the frequency count K.

The embedding $z(x) = B^{1/2} u_\omega(x)$ serves as both query and key representation. Define the distance score S and the cross-kernel matrix G_{*X} by

$$\mathsf{S}_{qi} = -\frac{\|z(x_q^*) - z(x_i)\|_2}{T}, \qquad (G_{*X})_{qi} = e^{\mathsf{S}_{qi}}. \tag{21}$$

Replacing the dot-product score in (20) with S and using the normalized labels as values gives

$$W_{\mathrm{att}} = \mathrm{diag}(G_{*X}\mathbf{1})^{-1}G_{*X} = \mathrm{softmax}_{\mathrm{row}}(\mathsf{S}), \qquad \widehat{\widetilde{y}}_*^{\mathrm{NW}} = W_{\mathrm{att}}\widetilde{y}. \tag{22}$$

This is the Nadaraya–Watson kernel smoother [8, 17], whose weights are positive and sum to one for each query.

The ridge predictor uses the same query–key similarities G_{*X} but aggregates the fitted coefficients α of (18), with no row normalization:

$$\widehat{\widetilde{y}}_*^{\mathrm{ridge}} = b\mathbf{1}_* + G_{*X}\underbrace{(G + n\lambda I)^{-1}(\widetilde{y} - b\mathbf{1})}_{\alpha}. \tag{23}$$

The solve combines the labels using the training Gram matrix G, accounting for overlap among kernel functions centered on the training observations. Writing the predictor in terms of the labels makes the difference from (22) explicit. For fixed geometry and λ, set

$$c = C_\lambda^{-1}\mathbf{1}, \qquad s = \mathbf{1}^\top c, \qquad P = C_\lambda^{-1} - \frac{cc^\top}{s}. \tag{24}$$

Substitution of (18) yields

$$\widehat{\widetilde{y}}_*^{\mathrm{ridge}} = W_{\mathrm{ridge}}\widetilde{y}, \qquad W_{\mathrm{ridge}} = G_{*X}P + \frac{\mathbf{1}_* c^\top}{s}, \qquad W_{\mathrm{ridge}}\mathbf{1} = \mathbf{1}_*, \tag{25}$$

where the last identity follows from $P\mathbf{1} = 0$. The unpenalized intercept makes the effective ridge weights sum to one, as the smoother's weights do, but individual ridge weights can be negative. These label weights differ from the expansion coefficients α, which sum to zero. The identity holds for fixed geometry and before output clipping; because the geometry is itself learned from the labels, the full procedure is not linear in them.

5.4 Categorical inputs and binary responses

A numerical Fourier map assigns meaning to input differences. Applying it directly to arbitrary category codes would make the distance depend on the numbering of those categories. For a categorical field with c levels observed in the fitting pool, let $e_v \in \mathbb{R}^c$ be its indicator vector and let $\Phi \in \mathbb{R}^{c\times(c-1)}$ have orthonormal columns perpendicular to $\mathbf{1}$. The categorical coordinates are $\Phi^\top e_v$. This removes the constant direction of a full indicator encoding. A previously unseen level maps to zero, the centroid of the fitted level coordinates; a missing category is represented as an explicit level when present in training.

For mixed inputs, the representation concatenates the numerical Fourier pairs with these fixed categorical coordinates:

$$u(x) = \mathrm{col}(u_\omega(x_{\mathrm{num}}), \Phi_1^\top e_{v_1}, \ldots, \Phi_m^\top e_{v_m}). \tag{26}$$

Here m is the number of categorical fields. One full trace-one metric acts on the concatenated representation, so the categorical fields can participate in cross terms with numerical features. The matrix-exponential parameterization and Laplace kernel are unchanged. Pair amplitudes in (9) apply to the numerical Fourier blocks; a categorical field's trace mass is the trace of its corresponding principal block. These blocks together account for the entire metric trace.

A binary target can be fitted with the same squared-error geometry objective and ridge head. For $y \in \{0, 1\}$, the population squared-error minimizer is the conditional probability, although a

fitted ridge score $s(x)$ need not lie in $[0,1]$. The classification experiments therefore use $s(x)$ for ranking and fit a sigmoid on a separate calibration set $\mathcal{C}$:

$$p(x) = \{1 + \exp[-(as(x) + b_c)]\}^{-1}, \tag{27}$$

$$(\widehat{a}, \widehat{b}_c) = \arg\min_{a,b_c} \sum_{i \in \mathcal{C}} \{-y_i \log p(x_i) - (1 - y_i) \log[1 - p(x_i)]\} + \frac{a^2}{2\mathcal{R}}, \qquad \mathcal{R} = 10^6. \tag{28}$$

The implementation uses logistic regression with an unpenalized intercept and weak slope regularization. Calibration rows are excluded from geometry training, ridge fitting and model selection. The same calibration procedure is applied to CatBoost's raw classification scores. Positive fitted slopes preserve score rankings; all reported calibration fits have positive slopes.

6 California Housing experiment

6.1 Data and selection protocol

California Housing contains 20,640 observations with eight numerical inputs; its source is the housing data studied by Pace and Barry [9]. We use the scikit-learn representation, with targets in units of \$100,000. For seeds $0, 1, 2$, random 80/20 splits yield 16,512 outer-training rows and 4,128 test rows. An inner 80/20 split, using random state $1000 + s$, gives 13,209 fitting rows and 3,303 validation rows. Every comparator uses the same saved row indices. Input standardization and target mean and standard deviation are fitted only on the inner-fitting rows.

The selected spectral model has one bank, four frequencies per feature and $p = 64$ coordinates. It has 2,079 metric coefficients, 32 frequency-gap coefficients and one log-bandwidth: 2,112 geometry parameters. Training adds one episodic diagonal-shift parameter, referred to as noise in the implementation. Neither tree predictions nor tree-derived features enter this model.

The starting checkpoint comes from an earlier eight-configuration validation search. Each selected checkpoint used standardization, four frequencies initialized on a geometric grid between 0.025 and 0.5, and 300 training steps with 384 support and 256 query rows. Refinement starts from that saved checkpoint and uses 1,536 support and 512 query rows for up to 800 further steps. The recipe was chosen using seed zero's validation set and fixed for the remaining seeds. Appendix A gives optimizer details.

Checkpoints are compared every 200 steps using 2,048-landmark validation regression. For the selected checkpoint, an exact head chooses λ from the nine-value grid $\{10^{-7+\ell/2} : \ell = 0, \ldots, 8\}$ and chooses whether to clip predictions to the fitting-label range. All three splits select step 800 and clipping. Their chosen λ values are 10^{-6}, $10^{-5.5}$ and 10^{-6}.

After freezing those selections, each final run restarts from the same original checkpoint and performs the selected 800 refinement steps on all outer-training rows. The inner-fitted transformations are retained to preserve the warm-start coordinates. The exact head is then fitted on all outer-training rows. All three geometry checkpoints were saved before the new test evaluation. A separate ablation freezes the inner-refined geometry and refits only the head.

LightGBM [7] and CatBoost [10] each receive a 16-configuration Sobol validation search, a 4,000-iteration cap and early stopping with patience 150. Each selected tree model is refitted on all outer-training rows for its selected iteration count. Search and computation budgets differ across methods. Earlier exploratory work also used these test splits, so they are not untouched holdouts.

6.2 Results

Table 2 reports mean and sample standard deviation across splits for RMSE and the coefficient of determination R^2. The refined model with full geometry refitting has mean RMSE \$820 below the retained tuned CatBoost baseline, a relative reduction of 1.90%. It has lower RMSE on each of the three matched splits (Table 3).

Table 2: California Housing test performance. All spectral rows use a single bank with four frequencies per feature. Standard deviations describe three overlapping random splits.

Model	RMSE (dollars)	R^2
Original spectral, Nyström (2,048)	$46{,}517 \pm 1{,}380$	$.8361 \pm .0084$
Original spectral, exact head	$44{,}816 \pm 1{,}477$	$.8479 \pm .0088$
Refined spectral, frozen geometry	$42{,}444 \pm 1{,}112$	$.8636 \pm .0055$
Refined spectral, full refit	$\mathbf{42{,}366} \pm 1{,}211$	$\mathbf{.8641} \pm .0060$
Tuned LightGBM	$43{,}568 \pm 488$	$.8563 \pm .0026$
Tuned CatBoost	$43{,}186 \pm 932$	$.8588 \pm .0050$

Table 3: Matched test RMSE in dollars. "Frozen" holds the inner-refined geometry fixed while fitting the exact head on all outer-training rows.

Split seed	Spectral full refit	Spectral frozen	CatBoost
0	41,004	41,183	42,111
1	42,771	42,863	43,772
2	43,323	43,286	43,674

Replacing the initial approximate decoder with an exact head reduces mean RMSE by about \$1,700 without changing that geometry. The exact head reselects its regularizer and clipping on validation, so this difference reflects the full decoder change, including that reselection. Further geometry refinement reduces error by roughly another \$2,372 with geometry held fixed during the outer refit. That refinement changes support size, learning rates and optimization budget together. Refining geometry on the full outer-training pool provides a smaller additional mean reduction of about \$78. This last step does not help on every split: on split seed 2 the frozen geometry has RMSE \$43,286 against \$43,323 for the full refit (Table 3).

The full-refit margins over CatBoost on the three splits are \$1,107, \$1,001 and \$351. The three overlapping splits provide limited evidence about performance differences across independent samples. Results on other datasets are reported in Section 7.

6.3 The fitted representation

Figure 1 shows the full-refit model for split zero, specified by the split index rather than selected for its geometry. The metric assigns approximately 44.96% of its trace to latitude and 31.48% to longitude, a combined 76.44%. This allocation describes the Fourier-space distance rather than a share of predictive accuracy, and the local sensitivity in (13) also depends on the frequencies and cross terms.

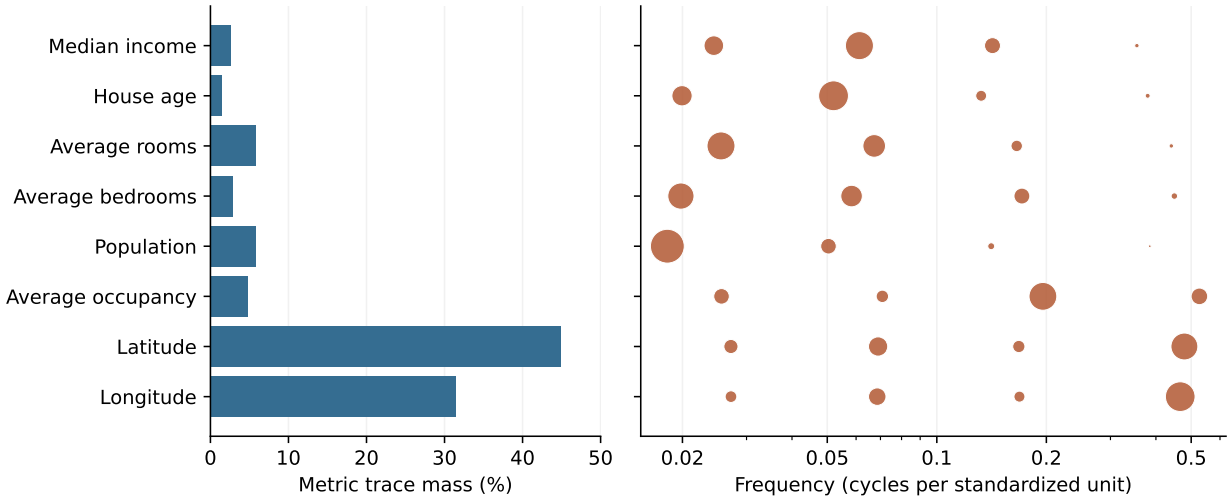


Figure 1: Saved single-bank geometry for split zero after the full refit. Left: each feature's metric trace mass ρ_j. Right: its four learned frequencies in cycles per standardized input unit; marker area is proportional to the within-feature mass a_{jk}^2. The panels are computed from the audited checkpoint diagnostics without additional fitting.

6.4 Computation and numerical checks

The symmetric matrix exponential is evaluated through

$$R(A) = \frac{\exp(A/2)}{\|\exp(A/2)\|_F}, \qquad RR^\top = B(A), \qquad z(x) = R(A)^\top u_\omega(x). \tag{29}$$

The identity is exact, and no truncated or low-rank approximation of B is used. Appendix B records its derivative structure for later computational work.

A dense matrix exponential costs $O(p^3)$ arithmetic. Transforming n embeddings costs $O(np^2)$; forming all pairwise distances costs $O(n^2p)$. Exact ridge factorization costs $O(n^3)$ and requires $O(n^2)$ storage. These are separate computational costs: reducing the ridge support size does not simplify the matrix exponential. At $n = 16{,}512$, one float64 Gram matrix occupies about 2.18 GB (2.03 GiB), before factors, cross-kernels and workspace.

Once the geometry is frozen, $R(A)$ and the training embeddings can be cached. Prediction then requires Fourier evaluation, the fixed linear transform and kernel evaluations, with no new matrix exponential. During training, changing A requires updating the exponential and differentiating through it. The reported timings do not isolate this operation.

On the NVIDIA GB10 GPU in float64, mean additional geometry-selection time was 165.9 seconds per split, full geometry refitting took 136.0 seconds, and exact final head fitting plus prediction took 8.9 seconds. These figures exclude the initialization search and exploratory candidates, so they are not end-to-end training costs.

The saved audit recomputes metrics from predictions, checks identical split indices, verifies the canonical constraints and inverse and checks relative exact-solve residuals below 10^{-9}. Source snapshots, selection records and predictions accompany the results.

7 Additional regression and classification experiments

Hourly Bike Sharing, Taiwan credit default and Adult income test the same single-bank construction beyond California Housing [4, 21, 2]. The mixed-input representation and binary-response readout are specified in Section 5.4. These runs train geometry from scratch; they do not reuse the California geometry.

7.1 Data and evaluation protocol

The hourly Bike Sharing file contains 17,379 observations. Its `casual` and `registered` columns sum to the target and are excluded, together with the row identifier and date string. The retained predictors describe calendar time and observed weather. The first 60% of dates provide 10,389 fitting rows, the next 20% provide 3,502 validation rows and the final dates provide 3,488 test rows. Training ends on March 13, 2012, and testing begins on August 7, 2012. Three optimizer seeds use this same chronology. The weather predictors are observed at the target hour, so the task is contemporaneous regression; forecasting under unknown future weather is not evaluated.

Taiwan contains 30,000 observations with 23 predictors and a binary next-month default label. Three stratified splits each allocate 19,200 rows to fitting, 2,400 to calibration, 2,400 to validation and 6,000 to testing. The numerical amounts retain their original values. Sex, education and marital status are treated as categorical fields.

Adult predicts whether annual income exceeds $50,000 from 14 census attributes. The official 16,281-row test partition is retained. For each of three seeds, the original 32,561 training rows are divided into 26,048 fitting, 3,256 calibration and 3,257 validation rows. Missing categories are represented explicitly and all rows are retained. The sampling-weight attribute is used as an input feature; the reported metrics are unweighted.

Both model families use identical row assignments. Calibration and validation observations remain outside the final model fit; these added experiments do not perform the outer-pool refit used for California. ROC AUC, the area under the receiver operating characteristic curve, selects classification models. A separate sigmoid fit on calibration observations supplies probabilities for log loss, Brier score and accuracy. RMSE selects regression models; regression results also report mean absolute error (MAE).

7.2 Training and selection

The spectral representation has four frequencies per numerical feature. Fixed categorical contrasts are appended before applying the full metric. Numerical standardization and a quantile-to-normal transformation are compared on seed zero. Its selected preprocessing recipe is then fixed for seeds one and two. Each model trains for up to 800 steps using 1,024 support rows and 256 disjoint query rows. Checkpoints at steps 0, 200, 400, 600 and 800 are compared with a 512-landmark head. The selected checkpoint is evaluated with 2,048 landmarks over the nine-value ridge grid used for California. Exact validation solves compare the resulting ridge and its adjacent grid values. This restricted search limits the number of dense factorizations.

Adam begins at learning rate 0.01 for diagonal metric coordinates, log bandwidth and log noise, and 0.002 for off-diagonal coordinates and log-frequency gaps. Rates decay to 20% of these values by the cosine schedule. The episode penalty, parameter bounds and gradient clipping follow the earlier construction. Target standardization is fitted on fitting rows and is also used for binary-label squared-error training.

CatBoost receives 16 Sobol configurations on seed zero, at most 3,000 trees per configuration and early stopping with patience 150. The selected configuration is fixed for the other two seeds,

which retain their own stopping iterations. CatBoost uses native categorical processing. Every spectral and CatBoost selection is saved before any test prediction; the search budgets differ between families.

7.3 Results

Table 4: Hourly Bike Sharing test results on the fixed chronological holdout. Standard deviations describe three training runs on the same test rows.

Model	RMSE	MAE	R^2
Spectral kernel	92.61 ± 1.25	64.23 ± 0.88	0.8236 ± 0.0047
Tuned CatBoost	107.45 ± 6.87	75.78 ± 5.91	0.7619 ± 0.0298

Table 5: Binary classification test results after separate sigmoid calibration. Larger AUC and smaller log loss or Brier score indicate better performance. Adult runs share the official test partition.

Dataset	Model	AUC	Log loss	Brier	Accuracy
Taiwan	Spectral	0.7873 ± 0.0021	0.4277 ± 0.0024	0.1339 ± 0.0009	0.8213 ± 0.0025
Taiwan	CatBoost	0.7828 ± 0.0027	0.4276 ± 0.0014	0.1340 ± 0.0005	0.8213 ± 0.0013
Adult	Spectral	0.9147 ± 0.0005	0.3153 ± 0.0017	0.0983 ± 0.0006	0.8620 ± 0.0008
Adult	CatBoost	0.9274 ± 0.0005	0.2775 ± 0.0008	0.0880 ± 0.0002	0.8736 ± 0.0007

Tables 4 and 5 report every added dataset under the frozen protocol. On Bike Sharing the spectral RMSE is lower on each seed, ranging from 91.3 to 93.7 against 99.5 to 111.7 for CatBoost. On Taiwan the paired AUC differences, spectral minus CatBoost, are 0.0064, 0.0044 and 0.0028, and the calibrated log losses differ by at most 0.0011 on any split.

Variation across the three Bike Sharing runs reflects training randomness on one chronology, and Adult varies the fitting and calibration partitions while retaining one official test set. Those standard deviations therefore do not estimate uncertainty across independent test samples.

The independent audit verifies all 18 model runs, including source-file hashes, split identity, validation selections, calibration separation, positive metric eigenvalues, ridge residuals and prediction replay. The accompanying source, per-run records and audits are listed in Appendix C.

8 Discussion

The fitted geometry can be examined through its frequencies, trace allocation and cross terms. Their effects depend on one another: local sensitivity depends jointly on frequency and metric weight, as shown in (13). The canonical parameterization fixes the scale conventions needed to compare these quantities across fitted models.

Computation depends on two dimensions. The dense matrix exponential grows cubically with representation dimension, while the exact ridge solve grows cubically with training size. The California model uses a 64-dimensional metric, but a compact embedding can still produce a full-rank Gram matrix. Approximations to the geometry and to the ridge fit therefore need separate evaluation. In particular, the California decoder comparison shows that the landmark count can materially affect prediction even when the geometry is held fixed.

The California holdouts were reused during exploratory development, and the selected refinement recipe benefited from earlier searches. The additional datasets use the frozen protocol of Section 7.

Geographic blocking for California, further datasets, comparisons under matched budgets and component ablations remain open. Comparisons with fixed frequencies, diagonal metrics and Gaussian kernels are needed to isolate the contribution of each component.

The added datasets distinguish ranking from probability quality. On Taiwan, the spectral AUC exceeds CatBoost's by 0.003 to 0.006 on each split, while the calibrated log losses agree to within 0.0011. The Bike Sharing comparison rests on one time boundary with observed weather.

On Adult the single-bank model has lower predictive performance than CatBoost: its mean test AUC is 0.9147 against 0.9274, and its calibrated log loss is 0.3153 against 0.2775. Each reported fit uses one spectral map, one metric and one bandwidth across its training population. Whether the Adult gap comes from this shared geometry, preprocessing, the training objective or incomplete optimization remains open.

9 Conclusion

This paper develops a kernel regression model in which the prediction task determines the representation of similarity. It uses Fourier frequencies to represent variation at different input scales, a metric to weight and couple those coordinates and a Laplace kernel to convert distance into similarity. Trace normalization and the matrix-exponential parameterization remove compensating scales, so amplitudes and feature masses can be recovered from the fitted metric without introducing redundant parameters.

The single-bank results are dataset-dependent. The spectral model has lower regression error than the tuned tree baselines on the reported California splits and Bike Sharing chronology. Taiwan shows slightly higher AUC with nearly equal calibrated log loss; Adult favors CatBoost on both measures. Reused California holdouts and unequal tuning budgets limit the comparison. The decoder experiments also show that a compact learned representation does not by itself make an approximate ridge fit adequate. Scaling this approach requires reducing the cost of the readout while retaining the predictive benefit of the learned geometry.

A longer-term direction is a tabular foundation model that learns a shared representation across datasets and adapts to a new table through its labeled context. That extension requires a representation that accommodates different feature schemas and a training objective spanning tasks. For large context sets, it also requires an alternative to the dense exact solve. The present single-task models provide a baseline for evaluating those changes. The question for subsequent work is whether spectral geometry learned across tasks can improve prediction on unseen tables while reducing the amount of task-specific fitting.

A Optimizer coordinates and refinement settings

For diagonal coordinates, use the $p-1$ Helmert vectors h_r, $r = 1, \dots, p-1$: the first r entries equal $1/\sqrt{r(r+1)}$, entry $r+1$ equals $-r/\sqrt{r(r+1)}$, and the rest are zero. Together with symmetric off-diagonal basis elements,

$$A = \sum_{r=1}^{p-1} \beta_r \operatorname{diag}(h_r) + \sum_{a<b} \gamma_{ab} \frac{e_a e_b^\top + e_b e_a^\top}{\sqrt{2}}. \tag{30}$$

This basis is orthonormal under the Frobenius inner product and has no free trace direction. For $p = 64$, it contains $63 + 2{,}016 = 2{,}079$ coefficients.

The initial search combines standard or quantile preprocessing, $K \in \{4, 8\}$, and two paired settings of learning rate and initial maximum frequency: $(0.02, 0.5)$ and $(0.04, 1.5)$. All three selected

initial models use standardization, $K = 4$, and the first pair. Initial validation uses 512 landmarks; a subsequent rank selection compares 512, 1,024 and 2,048, selecting 2,048 on every split.

Refinement uses Adam with initial learning rate 0.003 for diagonal metric coefficients, τ and log-noise, and 0.0006 for off-diagonal coefficients and frequency gaps. At step t, both rates are multiplied by $0.2 + 0.4[1 + \cos(\pi t/800)]$. The gradient norm is clipped at 10. Log-gaps are bounded to $[-9, 2]$, τ to $[-5, 3]$, and noise to $[10^{-4}, 1]$, initialized at 0.1. If the eigenvalue spread of A exceeds 20, its coordinates are uniformly rescaled to spread 20. Since $\kappa(B) = \exp(\lambda_{\max}(A) - \lambda_{\min}(A))$, this bounds the metric condition number by e^{20}. Checkpoint validation uses the same nine-value ridge grid as exact-head selection.

B Matrix-exponential differentiation

For a symmetric perturbation E, define the Fréchet derivative

$$L_A[E] = \frac{d}{dt} \exp(A + tE)\Big|_{t=0} = \int_0^1 \exp((1-s)A) E \exp(sA)\, ds. \tag{31}$$

Numerical methods for this derivative are studied by Al-Mohy and Higham [1]. Applying the quotient rule to (11) gives

$$DB(A)[E] = \frac{L_A[E] - B(A) \operatorname{tr} L_A[E]}{\operatorname{tr} \exp(A)}. \tag{32}$$

Its trace is zero, as required for a perturbation along the unit-trace surface. If $A = V \operatorname{diag}(a) V^\top$, the equivalent expression is

$$L_A[E] = V(F \odot (V^\top E V))V^\top, \qquad F_{rs} = \begin{cases} (e^{a_r} - e^{a_s})/(a_r - a_s), & a_r \neq a_s, \\ e^{a_r}, & a_r = a_s. \end{cases} \tag{33}$$

The repeated-eigenvalue case uses the continuous divided-difference limit. The implementation computes and differentiates the dense exponential by automatic differentiation; no faster matrix-function approximation is evaluated.

C Code and reproducibility

The implementation, recorded checkpoints, prediction arrays and manuscript are available at https://github.com/asudjianto-xml/Tabular-Spectral-Kernel. Release `v0.1.0` contains the single-bank experiments reported here. Installation instructions and the distinction between verification, replay and training are given in `README.md` and `REPRODUCIBILITY.md`.

The package accepts data and output directories from the command line. Public datasets are downloaded into the specified cache; no author-specific directory is required. A small synthetic CPU run checks differentiation through the matrix exponential and ridge solve. For example:

```
tsk smoke --device cpu --output runs/install-check
tsk verify --output runs/recorded-audit
tsk download --dataset all --data-dir data
tsk reproduce --dataset bike --mode replay --seeds 0 \
  --device auto --data-dir data --output runs/bike-replay
```

Verification recomputes per-run metrics from the bundled predictions, checks split separation and artifact hashes and exports aggregated results as a CSV file. The `assets` command regenerates the learned-geometry figure. Replay loads the recorded geometry and rebuilds its exact ridge head from the public data, comparing predictions with the saved arrays. Training repeats the documented fitting protocol and can produce different results across numerical platforms.

Source files are under `src/tabular_spectral_kernel/`, with the experiment implementations in `_paper/` and the saved evidence in `recorded/`. California training uses the published inner-training warm starts by default. Its initialization search can also be rerun, but the historical adaptive refinement screen is not reconstructed: the release retains the selected single-bank recipe. The additional-task commands repeat preprocessing selection, geometry fitting, ridge selection, tree tuning and calibration before test evaluation.

The recorded environment uses PyTorch 2.13.0+cu130, NumPy 1.26.4, scikit-learn 1.6.1, CatBoost 1.2.10 and LightGBM 4.7.0. The repository records these versions separately from the supported installation dependencies. CPU execution is available, but the dense exact heads still require quadratic memory and cubic factorization work. Reference metadata and the claims checked against primary sources are recorded in `paper/identifiable_spectral_reference_audit.json`.

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